COMPACT BENCHMARK BRIEF / FRESH RECOVERY

SkillRet on CPU

A separately authored, abridged report from the validated benchmark metrics. Primary quality: the same 32 queries, all 6,006 skills. Warm latency: a separate shared 32 queries x 2 repetitions.

MRR@5 averages 1/rank for the first relevant result within five, or zero if absent. Recall@5 measures the share of all judged-relevant skills found; Hit@1 tests the first result; Complete@5 requires the full gold set.

Best observed full-catalog quality: Gemma INT8 hybrid, with MRR@5 0.883 and Recall@5 77.1%. This is a descriptive result on this cohort, not a general model ranking.

Lower-latency hybrid option: BGE-small hybrid recorded median 99.7 ms, versus 509.4 ms for the highest-MRR pipeline. Choose according to the deployment quality/latency budget; these CPU measurements are not service-level guarantees.

E5-small hybrid is a useful quality/latency compromise in this sample: MRR@5 0.875 at 106.2 ms. Its point estimate is only 0.008 below the highest-MRR pipeline; that small gap should not be treated as broad superiority.

Matched full-catalog quality

Pipeline	MRR@5	Recall@5	Hit@1	Complete@5
BM25	0.740	60.4%	71.9%	43.8%
BGE-small hybrid	0.839	70.8%	78.1%	53.1%
E5-small hybrid	0.875	74.0%	84.4%	56.2%
Gemma INT8 hybrid	0.883	77.1%	84.4%	62.5%
BGE shortlist + GLiNER2.5	0.204	42.2%	6.2%	31.2%
BGE shortlist + Verdict	0.171	19.8%	9.4%	9.4%
BGE shortlist + Laya multilingual	0.119	19.3%	6.2%	12.5%
BGE shortlist + Julia-1	0.103	17.2%	0.0%	6.2%

The native rows all reorder the same BGE+BM25 hybrid shortlist. Its initial MRR@5 is 0.839 and candidate-any-hit ceiling is 0.906. Missing first-stage gold skills cannot be recovered by any second-stage ranker.

None of the four binary pairwise adapters improves MRR@5 over that initial ranking in this tested setup. This does not establish that native short-label choice routing is generally poor.

For a Jev-like router, treat native listwise choice over short skill labels as a different, untested pipeline. Twenty complete discovery headers exceed some native head budgets; the tested pairwise prompt is not a substitute for that design.

Catalogs and coverage

All values here use the same fresh 32-query population. Small catalogs are per-query, deterministic, nested and gold-preserving; they are not a single global small index. Retrieval-only 64-query measurements are preserved separately; the primary comparison filters them to the same selected native-query IDs.

MRR@5 across the four controlled catalogs

Pipeline	10 skills	100 skills	1,000 skills	6,006 skills
BM25	0.901	0.859	0.799	0.740
BGE-small hybrid	0.975	0.922	0.912	0.839
E5-small hybrid	0.984	0.945	0.922	0.875
Gemma INT8 hybrid	1.000	0.953	0.922	0.883
BGE shortlist + GLiNER2.5	0.608	0.367	0.259	0.204
BGE shortlist + Verdict	0.504	0.294	0.179	0.171
BGE shortlist + Laya multilingual	0.324	0.164	0.094	0.119
BGE shortlist + Julia-1	0.381	0.199	0.152	0.103

Rerankers receive up to 20 BGE hybrid candidates, or all 10 candidates in the smallest condition. Higher scores on smaller, gold-preserving catalogs should not be read as a production scaling forecast.

Multi-positive queries: 15 of 32

Pipeline	Recall@5	Complete@5
BM25	48.9%	13.3%
BGE-small hybrid	57.8%	20.0%
E5-small hybrid	64.4%	26.7%
Gemma INT8 hybrid	64.4%	33.3%
BGE shortlist + GLiNER2.5	30.0%	6.7%
BGE shortlist + Verdict	22.2%	0.0%
BGE shortlist + Laya multilingual	14.4%	0.0%
BGE shortlist + Julia-1	23.3%	0.0%

Recall retains every judged-positive skill in the denominator. Complete@5 requires all gold skills, so finding one relevant skill can still be an incomplete answer.

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Warm CPU request cost

Identical first 32 frozen query IDs, two fresh observations each, full 6,006-skill catalog. Lower latency is better; p95 describes this measured sample only.

End-to-end latency and native adapter component

Pipeline	Median total ms	Observed p95 ms	Median native ms
BM25	6.4	10.5	n/a
BGE-small hybrid	99.7	160.6	n/a
E5-small hybrid	106.2	160.0	n/a
Gemma INT8 hybrid	509.4	925.3	n/a
BGE shortlist + GLiNER2.5	7,357.4	18,550.1	7,238.4
BGE shortlist + Verdict	17,349.1	23,084.6	17,199.7
BGE shortlist + Laya multilingual	14,682.0	34,154.2	14,570.1
BGE shortlist + Julia-1	4,335.4	11,813.8	4,231.8

Native totals include fresh BGE/BM25/RRF retrieval plus reference-adapter token preflight, query fitting and scoring. Timed requests do not reuse score caches; document vectors/indexes are prebuilt. These are pipeline times, not isolated model-forward times.

BGE uses each native environment: Transformers 5.0.0 (Julia), 4.57.6 (GLiNER), 5.18 (others). Fresh shortlist equality is asserted. Stage medians need not sum to the total median.

Recorded process peaks, with stage context

Model	Recorded stage	Peak process MiB	CPU threads
BM25	Corpus build + retrieval	92	4
BGE-small	Corpus build + retrieval	907	4
E5-small	Corpus build + retrieval	1,372	4
Gemma INT8	Corpus build + retrieval	3,305	4
GLiNER2.5	32-query pair scoring	2,248	4
Verdict	32-query pair scoring	1,826	4
Laya multilingual	32-query pair scoring	2,821	4
Julia-1	32-query pair scoring	1,685	4

Recorded CPU: INTEL(R) XEON(R) PLATINUM 8573C. These are mixed-stage process peaks, not model-weight or serving-only memory. Retrieval includes batch-16 corpus encoding; native rows are quality-stage pair scoring. Native timing processes separately hold BGE plus the classifier. Setup is excluded from warm latency.

Limits and methods

The companion archive retains full tables, fresh raw data, source revisions and the executed notebook.

Native MRR change versus the same-query initial shortlist

Adapter	Paired MRR change	Descriptive 95% interval
GLiNER2.5	-0.634	[-0.747, -0.513]
Julia-1	-0.736	[-0.840, -0.616]
Laya multilingual	-0.719	[-0.849, -0.562]
Verdict	-0.667	[-0.793, -0.531]

Paired query bootstrap: 2,000 resamples, seed 20261002; conditional on this protocol and its judgments. These pointwise intervals are not multiplicity-adjusted, causal or out-of-domain guarantees.

For context, BGE hybrid versus BM25 has estimated MRR change +0.099, interval [-0.016, +0.235]. This illustrates the limited precision of a 32-query comparison.

Distinct queries shortened: GLiNER2.5 0/32, Verdict 22/32, Laya multilingual 1/32, Julia-1 1/32. Embedding counts: BGE-small 2/64, E5-small 2/64, Gemma INT8 0/64 (all 64 encoded queries). Candidate cuts: 0 in 7568 unique model/query/candidate evaluations. Embedding document truncations: 0 across three runs of 6,006 documents. Native query prefixes may differ by candidate.

What was measured

- Primary quality: the same 32 outcome-independently selected queries. Runtime, not accuracy, determined the attempted extension.

- SkillRet v1.1 short headers: name plus the first 1,024 description characters, not full skill_md. No official full-text leaderboard reproduction.

- domain_vector uses the top-1 centroid of six skill major categories, then vectors within it. Existing skill metadata only; no gold-query routing or 13-tag classifier.

- Fixed pairwise binary prompts test reranking; native short-label listwise choice routing remains untested.

- Missing qrels mean non-relevant under released judgments, not that another skill cannot help. Coverage and truncation constrain results.

- Four CPU threads on one host. Recovery spinning-off ONNX timings are not pooled with historical default-ORT timings.

Historical evidence and the recovery boundary

Final scope is the common 32-query cohort. Only Julia/GLiNER completed 48; those results are supplements. A safety restriction involving a Microsoft telemetry connection blocked the 48-query Verdict stage; no inference retry occurred. Prospective privacy changes did not alter the measurements. Earlier 4,392-query values were genuinely measured and audited, but their original raw files became unavailable. They remain historical only. The earlier eight-query test had zero MRR headroom; neither historical set enters fresh means or intervals.

Pinned ThakiCloud/SKILLRET v1.1 dataset. Exact model revisions, fresh CSVs and full audit details are in the archive.

Analysis source SHA256: 1ab42bb6248d0076e3dbc2d7cb79d74f03f06ac8a4d760038a24161057a39ae2